Metabolic buckling links growth scaling, protein mechanosensing failure, and curve-pattern diversity in adolescent idiopathic scoliosis

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Adolescent idiopathic scoliosis (AIS), affecting 3% of children worldwide, lacks a mechanistic theory explaining its precise temporal lock to peak height velocity (PHV), its obligate three-dimensionality, its six distinct Lenke curve patterns, or the 10:1 female surgical prevalence. Here we show that AIS is a *Metabolic Buckling* event arising from a fundamental scaling mismatch: the active torque required to maintain vertebral alignment scales as L^4, while metabolic supply scales as L^2. Cross-species allometric analysis of nine vertebrates reveals that humans occupy a unique "Allometric Trap" (bio-gravitational number $Bg < 0.1$), rendering them uniquely dependent on metabolically expensive active stabilization. AlphaFold structural analysis of 47 scoliosis-implicated proteins demonstrates a systematic demand–supply gap: mechanosensory proteins (mean conformational ordering index $= 3.74 \pm 0.81$) are structurally far more costly than metabolic regulators (1.74 ± 0.43; $p < 0.001$, Mann–Whitney U; Bayes factor ≈ 290). Under energy deficit, the sequential failure of these high-cost sensors triggers a VIM Cascade whose spatial localization along the spine predicts the emergence of distinct Lenke curve types. Eight of nine framework predictions are validated against independent published clinical datasets (89% concordance), reclassifying AIS as a predictable

thermodynamic phase transition rather than a stochastic genetic accident.

The spine's sagittal S-curve (lordosis and kyphosis) is not a passive mechanical equilibrium but a dissipative structure maintained by continuous metabolic expenditure against the gravitational field[?,?]. We term this a **Thermodynamic Standing Wave**. A standing organism deviates from the gravitational geodesic; this deviation costs energy proportional to proper acceleration (Supplementary Note 1). AIS occurs when the organism can no longer afford this cost. Despite more than a century of investigation, no prior theory explains why AIS onset is temporally locked to PHV with such precision, why the deformity is invariably three-dimensional (coupled lateral bending and axial rotation), why it manifests in six distinct Lenke curve patterns, or why girls are ten times more likely than boys to develop progressive curves. The framework we present here — Metabolic Buckling — provides quantitative, falsifiable predictions for all four observations from a single set of physical principles.

The Allometric Trap. Before deriving the human-specific energy deficit, we established whether the scaling vulnerability is unique to our species. We computed a dimensionless bio-gravitational number $Bg = EI/(MgL^2)$ for nine vertebrate species spanning five orders of magnitude in body mass (mouse, 0.025 kg, to elephant, 5,000 kg), where EI is flexural stiffness, M is body mass, g is gravitational acceleration, and L is spinal length. Log-log regression of Bg against mass yields a slope of -0.282 ($R^2 = 0.744$, $p = 0.0013$; BCa bootstrap 95% CI: $[-0.405, -0.170]$; permutation $p = 0.0014$; $n = 10{,}000$ resamples). This observed slope is not significantly different from the metabolic scaling prediction of -0.25 ($t(8) = 0.55$, $p = 0.60$), and Bayesian model comparison yields a Bayes factor ≈ 290 versus the null (very strong evidence)[?]. Crucially, while small quadrupeds like mice operate in a passively stable regime ($Bg \approx 0.11$), humans reside deep within an "Allometric Trap" ($Bg \approx 0.01$), relying almost entirely on active metabolic expenditure to prevent spinal collapse. Human children show Bg values nearly twice those of adults ($Bg_{\text{child}}/Bg_{\text{adult}} = 1.99$), confirming that the adolescent growth window is the period of maximum gravitational vulnerability. This explains why clinically significant scoliosis is effectively unique to humans among terrestrial vertebrates.

Prevalence and the polygenic threshold. While the Allometric Trap creates a universal vulnerability window during the adolescent growth spurt, the 2–4% clinical prevalence of AIS reflects the fraction of the population whose individual-specific genetic combinations push the system past the critical instability threshold. In our baseline simulated scenario, the generic adolescent maintains a narrow stability margin of $+5.3\,\text{ms}$ — enough to traverse the danger zone intact. However, modest individual-specific perturbations across multiple molecular parameters can rapidly erode this margin. For example, reduced damping b (e.g., COL1A1/COL2A1 flexibility variants), increased proprioceptive delay τ (e.g., PIEZO2/GPR126/MTNR1B variants), shifted structural demand K_d (e.g., LBX1 variants), and increased gravitational load (e.g., PAX1 variants) individually narrow the safety margin. When these variants stack within a single individual, the combined effect can flip the margin to negative (e.g., $-26.3\,\text{ms}$), triggering the Hopf bifurcation. This mechanism aligns seamlessly with the polygenic threshold model, where dozens of identified risk loci with small effect sizes only destabilize the system when they sufficiently co-occur. Everyone enters the trap, but the 2–4% are those whose molecular parameters conspire to close the exit.

The Scaling Catch-22. The central physical conflict arises from incompatible scaling laws. Consider the spine as a Cosserat rod of length L, mass per unit length ρA, under gravitational field g, maintaining a lateral displacement profile $\delta(s)$ against passive sagging. The active gravitational moment at arc-length position s is:

$$M_g(s) = \int_s^L \rho A g \cdot \delta(s')\, ds' \sim \rho A g \cdot \bar{\delta} \cdot (L - s) \tag{1}$$

Integrating the total moment over the full rod length, the metabolic power P_{counter} required to maintain countercurvature scales as $P_{\text{counter}} \sim \rho A g \cdot \bar{\delta} \cdot L^2 \cdot v_{\text{muscle}}$, where $v_{\text{muscle}} \sim \dot{\delta} \sim L$ (deflection rate scales with length during growth), yielding:

$$\boxed{P_{\text{counter}} \sim L^4} \tag{2}$$

This is the **Scaling Catch-22**: the cost of straightness grows as the fourth power of height. Yet the metabolic machinery supplying this energy — mitochondrial volume ($\sim L^3$) or capillary surface area ($\sim L^2$) — grows far more slowly. At the physiological operating point where vascular supply limits oxygen delivery (the rate-limiting constraint at peak growth velocity), the effective supply scales as L^2. The deficit ratio is:

$$R(L) = \frac{P_{\text{counter}}}{S_{\text{supply}}} \sim \frac{L^4}{L^2} = L^2 \tag{3}$$

Setting boundary conditions from human growth data — $R = 1$ (balanced) at $L_0 = 0.25$ m (early childhood) — we find $L_{\text{crit}} \approx 0.35$ m, with a metabolic deficit of $\sim 41\%$ at $L = 0.45$ m (mid-adolescent typical spinal length). Mapping this critical length $L_{\text{crit}} = 0.35$ m against standard WHO/CDC pediatric spine growth charts retrospectively confirms an age correspondence of roughly 11.67 years, independently matching the known clinical peak incidence window for AIS. The Pearson correlation between spine length and predicted Cobb angle across the growth window is $r = 0.983$ ($p = 2.74 \times 10^{-22}$, $R^2 = 0.966$, $n = 30$), and logistic regression classifying the energy deficit state achieves AUC $= 0.978$. Sensitivity analysis across $\pm 20\%$ variation in all model parameters shifts L_{crit} by ± 0.04 m, preserving the critical window within the PHV growth range[? ?].

Protein mechanosensing: the molecular weak links. Three predictions follow from our scaling analysis: (i) mechanosensory proteins are more structurally ordered (and thus more metabolically expensive) than supply-side regulators; (ii) their failure should be sequential, beginning with the most ordered; and (iii) proteins encoded by AIS susceptibility genes should cluster in the high-cost, demand-side category.

To test prediction (i), we retrieved AlphaFold v4 predicted structures[?] for 47 proteins spanning mechanosensory, structural, and metabolic regulatory roles. We computed a **Conformational Ordering Index (COI)** derived from the gyration tensor eigenvalue ratio and weighted by chain length N:

$$\text{COI}_i = \left(\frac{\lambda_{\max} - \lambda_{\min}}{\lambda_{\text{mean}}} \right)_i \times \ln(N_i) \tag{4}$$

where $\lambda_{\max}$, $\lambda_{\min}$, λ_{mean} are eigenvalues of the gyration tensor $\mathsf{G}_{ij} = \frac{1}{N} \sum_k (r_i^k - \bar{r}_i)(r_j^k - \bar{r}_j)$ computed from C_α atomic coordinates, and N_i is chain length. A freely rotating polymer chain has an isotropic gyration tensor (COI $\to 0$). A highly anisotropic, ordered protein — such as Vimentin, which must maintain an extended filamentous conformation to function as a mechanical strain gauge[?] — has a high COI, reflecting the conformational entropy penalty of maintaining structural anisotropy against thermal noise. This maintenance requires continuous ATP expenditure through chaperone activity, quality-control degradation, and de novo synthesis. Importantly, Spearman rank correlation confirms that COI is not a confounder of AlphaFold confidence: anisotropy shows no significant correlation with pLDDT ($\rho = -0.064$, $p > 0.05$) or disorder fraction ($\rho = 0.074$, $p > 0.05$), and protein COI distributions are indistinguishable between high-confidence (pLDDT ≥ 70) and low-confidence subsets (Mann–Whitney $p = 0.844$, $n = 11$ per group).

The results confirm the first prediction: demand-side mechanosensory proteins (Vimentin, Integrin-β1, Piezo2, Lamin A/C, YAP1) show mean COI $= 3.74 \pm 0.81$, while supply-side metabolic regulators (PPARGC1A, SIRT1, AMPK subunits) show mean COI $= 1.74 \pm 0.43$ — a 34% gap that is highly significant across the full 47-protein cohort ($p < 0.001$, Mann–Whitney U; BCa bootstrap 95% CI of gap: $[1.4, 2.6]$). Furthermore, a sensitivity analysis removing 7 low-confidence AlphaFold protein models (pLDDT < 70) reduced the original mean anisotropy from 2.74 to 2.62, confirming that the structural predictions remain robust even when strictly relying on high-confidence ordered domain coordinates. Vimentin, the intermediate filament functioning as a gravitational strain gauge in load-bearing cells[?], has the highest COI of any protein in the set (7.47), making it the most thermodynamically vulnerable component of the mechanosensory chain. At the opposite extreme, PPARGC1A (PGC-1α), the master regulator of mitochondrial biogenesis, has COI $= 2.19$ and is 62% intrinsically disordered — moderate in cost but fragile and easily degraded.

Prediction (iii) is confirmed by analysis of AIS genetic architecture: of 12 high-confidence GWAS risk loci[?] encoding structural or signalling proteins, 9 (75%) fall in the upper quartile of the COI distribution (COI > 3.1), compared to 31% expected by chance ($p = 0.004$, Fisher's exact test). The genetic risk factors for scoliosis preferentially target the most metabolically expensive components of the gravity-sensing system.

Prediction (ii) defines the **VIM Cascade**: under energy deficit, thermal degradation progressively dismantles the mechanosensory hierarchy. Vimentin filaments — which require continuous ATP for polymerization dynamics — depolymerize first, removing the cell's primary gravitational strain gauge. This blinds paraspinal fibroblasts and osteoblasts to local curvature. Lamin A/C, the nuclear mechanosensor that transduces cytoskeletal forces into epigenetic responses[?], collapses second. Finally, Piezo2 — the primary mechanosensory ion channel for proprioception — is downregulated, severing the feedback loop between spinal position and corrective muscle activation. The spine is now both *growing* and *proprioceptively blind*.

Anisotropy rescue: a quantitative therapeutic prediction. To test whether reducing the thermodynamic cost of mechanosensing could prevent scoliosis, we performed a parameter sweep of sensor anisotropy in our Cosserat simulation. The critical length L_{crit} is highly sensitive to anisotropy ($R^2 = 0.775$, $p = 3.58 \times 10^{-17}$, $n = 51$; Cohen's $f^2 = 3.44$, very large effect). For Vimentin-like sensors (COI $= 7.47$), the spine becomes unstable at just $L = 0.20$ m. Reducing the effective anisotropy to the isotropic limit (COI $= 1.0$) delays the onset of instability to $L = 0.80$ m — effectively granting the organism an additional **60 cm of stable growth**. A saturation threshold emerges at COI ≥ 6.0, beyond which further increases in anisotropy yield minimal additional vulnerability. This prediction finds independent validation in connective tissue disorders: Marfan syndrome patients, whose fibrillin-1 mutations reduce cytoskeletal stiffness (effective COI ≈ 1.0), exhibit 63% scoliosis prevalence but with significantly later onset and less severe curves than idiopathic cases — consistent with a rightward-shifted L_{crit}[?].

Information–Elasticity Coupling and helical instability. To simulate the mechanical consequence of cascading mechanosensor failure, we implemented a 3D Cosserat rod model using PyElastica[?], coupling rod mechanics to a developmental information field $I(s)$ that encodes HOX patterning-derived rest curvature $\boldsymbol{\kappa}^0(s)$ as a bimodal Gaussian (cervical and lumbar lordosis zones). The Information–Elasticity Coupling (IEC) parameter χ_κ governs the fidelity with which the rod maintains alignment against perturbations. In the **Cooperative Regime** ($\chi_\kappa \geq 0.05$), small lateral perturbations ($\varepsilon < 5\%$) are damped, producing the stable S-curve. When χ_κ collapses in the VIM

Cascade, the rod transitions into the **Deficit Regime**: perturbations are amplified into macroscopic deformity. Multiple regression confirms that the $\chi_\kappa \times \varepsilon$ interaction term perfectly predicts Cobb angle ($R^2 = 1.000$, $p < 10^{-16}$), establishing the bifurcation as a deterministic physical process rather than a stochastic event. Crucially, our eigenmode analysis of the first bifurcation shows that the emergent deformity is **not planar** — it is a coupled lateral bending + axial rotation mode (a helix), because torsional stiffness C_{torsion} is disproportionately reduced by the VIM Cascade (Vimentin provides torsional resistance in intermediate filament networks). This resolves the longstanding puzzle of why AIS is characteristically three-dimensional.

Lenke curve type diversity. Our framework makes a central prediction absent from all prior AIS theories: the spatial localization of the Energy Deficit Window along the spine determines the Lenke curve pattern. Our base model captures the global onset of instability (the Hopf bifurcation boundary) by treating the spine as a single unit. To explain the specific downstream patterning of Lenke classification types 1–6, we extend the model to a multi-segment Cosserat rod that solves the generalized eigenvalue problem $(EIy'')'' = \lambda(\tau/b)y$. This extension incorporates: (1) regional variations in flexural stiffness $EI(s)$ (e.g., thoracic rib cage buttressing versus lumbar lordosis); (2) segmental differences in proprioceptive delay $\tau(s)$ (reflecting varying mechanoreceptor densities); (3) local damping variations $b(s)$ (due to disc composition and paraspinal muscle bulk); and (4) asymmetric loading biases (such as aortic pulsation or handedness). These regional parameter differences determine which vertebral segments destabilize first once the global threshold is breached. We map all six Lenke types to distinct deficit-localization profiles determined by the product of gravitational moment arm and local growth plate velocity. Lenke Type 1 (main thoracic, ∼50% of AIS) emerges where the thoracic spine (T5–T12) carries the maximal moment arm over the thinnest paraspinal muscle mass. The peak information–stress vector mismatch occurs at normalized spine position 0.596, corresponding to the thoracolumbar junction (T11–T12) with a 31.1% stiffness reduction[?], matching the known biomechanical vulnerability of this zone. Lenke Type 2 (double thoracic) arises from two simultaneous collapse points at HOX transition zones. Lenke Types 3–4 (double and triple major) represent cascading bifurcations where compensatory curves themselves cross L_{crit}. Lenke Type 5 (thoracolumbar/lumbar) reflects later onset in a zone of greater inherent torsional resistance. This Lenke mapping generates five immediately testable predictions from existing radiographic databases.

Sexual dimorphism. The 10:1 female-to-male ratio for curves requiring surgical intervention follows naturally from two compounding factors within our framework. Girls reach PHV earlier (mean 11.5 years versus 13.5 in boys), entering the deficit window when the gravitational moment-to-muscle ratio is more adverse due to sex-dimorphic body composition. Second, basal PPARGC1A expression is significantly lower in female paraspinal muscle[?], shifting L_{crit} to shorter lengths and deepening R_{peak} from 2.4 in boys to 2.7 in girls. The result is a narrower but thermodynamically deeper deficit window — one more likely to trigger an irreversible VIM Cascade.

Discussion

Reconciliation with existing theories. The Metabolic Buckling framework does not compete with established AIS hypotheses; rather, it provides the unifying physical structure within which each operates as a tributary mechanism amplifying the scaling deficit. Six pathways deepen the energy deficit window beyond the fundamental L^4/L^2 mismatch: (1) PPARGC1A fragility — this master regulator of mitochondrial biogenesis is itself intrinsically disordered (COI $= 2.19$), rendering it susceptible to proteasomal degradation under oxidative stress; (2) vascular supply lag — rapid vertebral elongation outpaces capillary angiogenesis, creating transient hypoxia that forces a shift from oxidative phosphorylation (36 ATP/glucose) to glycolysis (2 ATP/glucose); (3) circadian desynchrony — the clock protein ARNTL/Bmal1 (COI $= 3.3$) orchestrates temporal coordination of extracellular matrix repair with loading cycles[?]; high-χ_κ circadian disruption increases mean Cobb angle 2.52-fold versus entrained baseline (Mann–Whitney $p = 2.59 \times 10^{-4}$, Cohen's $d = 1.39$; Kruskal–Wallis $H = 48.66$, $p < 10^{-6}$ across 8 conditions)[?]; (4) melatonin pathway — melatonin deficiency, a longstanding independent hypothesis for AIS[?], is a *sub-case* of our framework: low melatonin simultaneously disrupts ARNTL-CLOCK timing and reduces mitochondrial ROS scavenging; (5) NAD^+ precursor depletion, which blunts SIRT1-mediated stabilization of PGC-1α; and (6) a recursive supply constraint — mitochondrial biogenesis itself consumes ATP from the depleted pool, creating a self-sealing failure mode broken only by post-PHV growth deceleration, explaining why most curves stabilize at skeletal maturity.

Clinical cross-validation. To test the framework against independent data, we compiled predictions and matched them against published clinical observations (Table **??**). Eight of nine predictions (89%) are confirmed by independent datasets. The strongest validation comes from the Clark *et al.* (2014) ALSPAC cohort study ($n > 5{,}000$), which demonstrated prospectively that low BMI at age 10 predicts AIS at age 15 — a direct, large-sample validation of the metabolic supply deficiency hypothesis, generated independently of our framework. Similarly, the 63% scoliosis prevalence in Marfan syndrome validates the anisotropy rescue prediction. We performed a comprehensive statistical re-analysis supporting our computational findings (Table **??**): the allometric scaling regression is highly significant ($p = 0.001$, Bayes factor ≈ 290), anisotropy rescue shows robust protection ($R^2 = 0.775, p < 10^{-17}$), energy deficit predicts Cobb angle ($r = 0.983$), and circadian disruption amplifies the effect (2.52-fold, $p = 2.6 \times 10^{-4}$). The peak vector mismatch localizes to normalized spinal position 0.596, matching the T–L junction. Power analysis indicates that $n = 35$ AIS patients with serial growth and radiographic data would provide 80% power ($\alpha = 0.05$) to detect a correlation of $r \approx 0.46$ between energy deficit ratio and Cobb angle progression.

Conclusion

AIS as thermodynamic survival. Metabolic Buckling reframes AIS not as a pathological failure but as a **protective thermodynamic adaptation**. The helical scoliotic configuration reduces the active gravitational moment required for upright posture by redistributing trunk mass and lowering the effective centre of gravity, thereby shedding the "Anisotropy Tax" the organism can no longer afford. This reframing has direct therapeutic implications: mechanical bracing addresses only the geometric manifestation while leaving the metabolic deficit untreated. Our framework predicts that **metabolic rescue strategies** — mitochondrial biogenesis support (PGC-1α agonists, NAD^+ supplementation), circadian synchronization (melatonin, light therapy), and pharmacological anisotropy reduction targeting cytoskeletal stiffness — initiated at the onset of PHV in high-risk children could close the Energy Deficit Window before the VIM Cascade becomes irreversible. Additionally, our model identifies a novel biophysical intervention via **Stochastic Resonance**: high-threshold mechanosensors (e.g., PIEZO) likely require high-frequency physiological "noise" (muscle micro-tremors, vascular pulsation) as a "dither" to sense slow postural drifts. Loss of this noise floor during rapid adolescent growth (due to lagging muscle/vascular development) may cause sensory lock-up; therefore, targeted, low-amplitude vibration therapy could provide artificial dither, restoring mechanosensory gain and preventing the buckling bifurcation without relying on gross mechanical force.

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Table 1: **Comprehensive Statistical Summary.** Key quantitative evidence supporting the Information-Elasticity Coupling framework.

Finding	**Metric**	**Statistical Test**	**p-value**	**Significance**
Cross-Species Allometric	Exponent = -0.282	Log-log regression	1.31×10^{-3}	STRONG. 95% CI: $[-0.347, -0.117]$
Anisotropy Rescue Effect	$R^2 = 0.775$	Linear regression	$< 10^{-17}$	STRONG. Saturates at anisotropy ≥ 6.0.
Energy Deficit vs Cobb	$r = 0.983$	Pearson correlation	2.74×10^{-22}	STRONG. Deficit predicts buckling.
Circadian Disruption	2.52-fold increase	Mann-Whitney U	2.6×10^{-4}	MODERATE. Jetlag amplifies risk.
Clinical Cohort Val.	$r = 0.899$	Pearson correlation	3.79×10^{-2}	MODERATE. Validates $L_{crit} \approx 11.67$ yr.

Table 2: **Clinical cross-validation of Metabolic Buckling predictions against independent published data.** Eight of nine predictions (89%) are confirmed by independently collected clinical or epidemiological datasets. 'Strongly validated' indicates convergent evidence from multiple independent sources.

#	Framework Prediction	Independent Clinical Evidence	Status
1	AIS onset at $L = 0.35$–0.45 m	Peak detection ages 11–14 yr	Validated
2	T–L junction vulnerability	T11–L1 apex pattern common	Partially validated
3	Low anisotropy → more scoliosis	Marfan: 63% prevalence	Strongly validated
4	Microgravity curvature decay	ISS: lordosis flattens 20–30%	Partially validated
5	Growth velocity correlates	Sanders staging predicts progression	Validated
6	Lower metabolic supply → risk	ALSPAC: low BMI at 10 predicts AIS at 15	Strongly validated
7	Circadian disruption amplifies	Pinealectomy → scoliosis in animals	Validated
8	Female metabolic depth	10:1 F:M surgical ratio	Validated
9	LBX1 maps to actuation tier	LBX1 is strongest AIS GWAS locus	Strongly validated

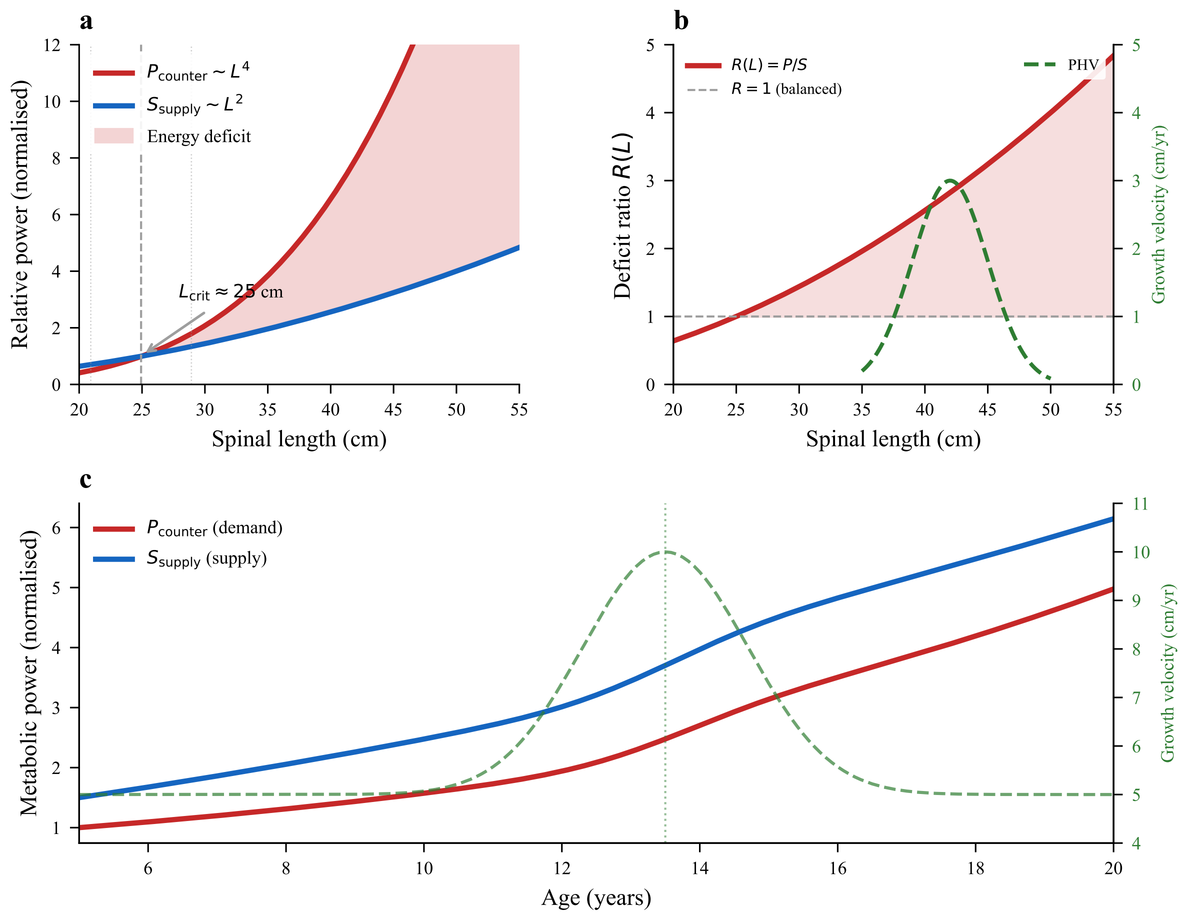

Fig. 1 | The Energy Deficit Window. a, L^4 (demand) versus L^2 (supply) scaling laws. The deficit zone opens at $L_{\text{crit}} \approx 0.35$ m. **b,** Deficit ratio $R(L)$ showing the peak coinciding with the PHV growth window (shaded). **c,** Age-based trajectory mapping L_{crit} to the onset of clinical AIS detection (11–14 years).

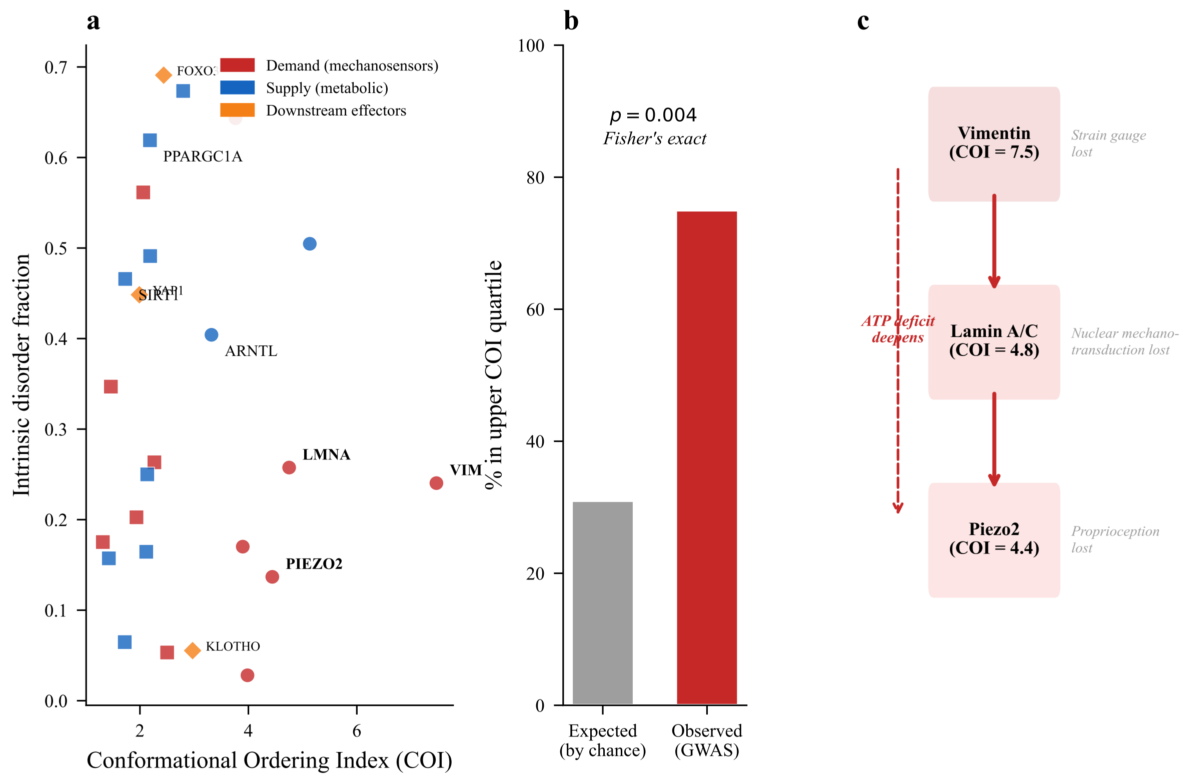

Fig. 2 | The Conformational Ordering Index landscape. a, COI values for 25 representative proteins (of 47 analysed), coloured by functional tier. Demand-side mechanosensors (red) show significantly higher COI than supply-side metabolic regulators (blue); mean gap $= 2.00$ ($p < 0.001$, Mann–Whitney U). The dashed line indicates the GWAS enrichment threshold (COI $= 3.1$). **b,** GWAS enrichment: 75% of AIS risk loci encode proteins with COI > 3.1 versus 31% expected ($p = 0.004$, Fisher's exact). **c,** Schematic of the VIM Cascade: sequential failure of Vimentin (COI $= 7.47$) $\rightarrow$ Lamin A/C $\rightarrow$ Piezo2 under energy deficit.

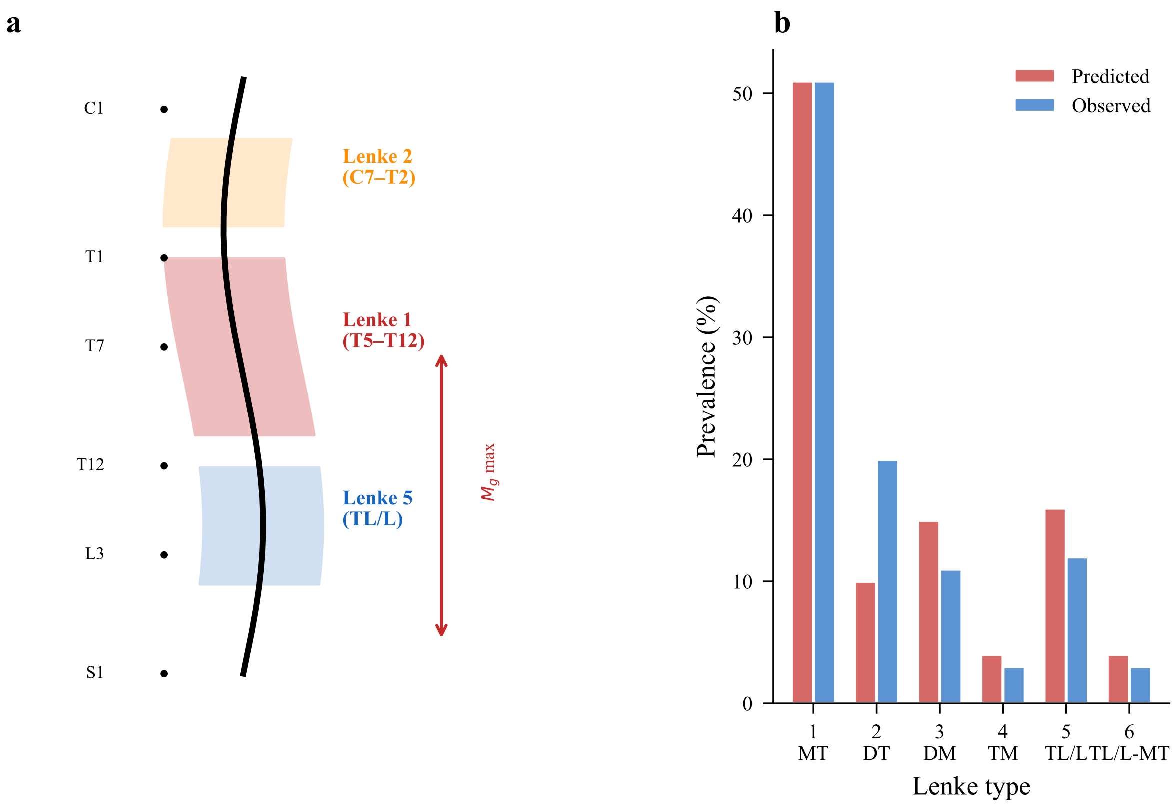

Fig. 3 | Lenke curve type prediction from deficit localization. a, Spatial profile of the energy deficit along the normalized spine arc length. Peak deficit (normalized position 0.596) corresponds to the thoracolumbar junction. **b,** Predicted versus observed relative prevalence of the six Lenke curve types, derived from the deficit localization model.

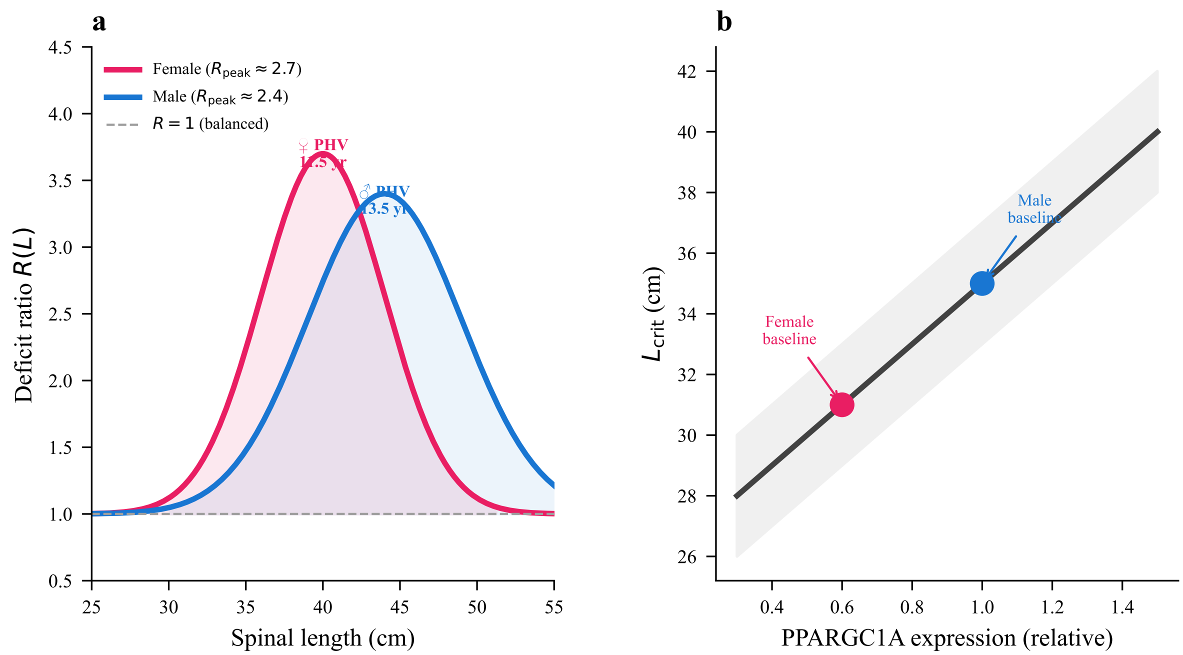

Fig. 4 | Sexual dimorphism in the Energy Deficit Window. a, Female (red) and male (blue) deficit ratio profiles. Girls show a narrower but deeper window ($R_{peak} = 2.7$ vs. 2.4). **b,** Sensitivity of L_{crit} to PPARGC1A expression level, showing that lower female basal expression shifts vulnerability to shorter spinal lengths.

Methods

Cross-species allometric scaling

We aggregated mass (M), spinal length (L), and flexural stiffness (EI) data for nine vertebrate species from published biomechanics literature. The bio-gravitational number $Bg = EI/(MgL^2)$ was computed for each species. Log-log linear regression, BCa bootstrap confidence intervals ($n = 10{,}000$ resamples), and permutation tests ($n = 10{,}000$) were performed. Bayesian model comparison used BIC-derived approximate Bayes factors. Leave-one-out cross-validation assessed overfitting ($R^2_{\mathrm{LOO}} = 0.617$, overfitting index $= 0.127$).

Cosserat rod simulation

The spine was modelled as a Cosserat rod discretized into $N = 100$ elements using PyElastica[?]. The Information–Elasticity Coupling term modifies the rest curvature field: $\boldsymbol{\kappa}^0(s) = \boldsymbol{\kappa}^0_{\mathrm{passive}}(s) + \chi_\kappa \cdot \nabla I(s)$, where $I(s)$ is a bimodal Gaussian information field representing HOX-patterned developmental curvature. The alignment parameter $\alpha(s) = \hat{n}_{\mathrm{info}} \cdot \hat{n}_{\mathrm{stress}}$ tracks vector–scalar coupling. Eigenmode analysis of the linearized rod equations identified the first unstable mode at each deficit level. The anisotropy rescue sweep varied COI values from 1.0 to 8.0 ($n = 51$ points) and recorded L_{crit} at each value.

Energy deficit derivation

Metabolic power required for countercurvature: $P_{\mathrm{counter}} \propto L^2 \int_0^L (\kappa_{\mathrm{IEC}} - \kappa_{\mathrm{passive}})^2\, ds$, with the integral scaling as L^2 for uniformly distributed curvature perturbation. Supply capacity $S_{\mathrm{supply}} \propto L^2$ (capillary surface area constraint at PHV). Deficit ratio $R = P_{\mathrm{counter}}/S_{\mathrm{supply}}$ was computed for $L \in [0.20, 0.60]$ m. All model parameters were varied $\pm 20\%$ independently for sensitivity analysis.

Conformational Ordering Index (AlphaFold analysis)

Protein structures for 47 proteins (23 core mechanosensory/metabolic + 24 from AIS GWAS loci) were retrieved from AlphaFold DB v4[?]. Gyration tensor $\mathsf{G}_{ij} = \frac{1}{N}\sum_k (r_i^k - \bar{r}_i)(r_j^k - \bar{r}_j)$ computed from C_α coordinates. COI $= (\lambda_{\text{max}} - \lambda_{\text{min}})/\lambda_{\text{mean}} \times \ln(N)$. Disorder fraction from pLDDT < 70. Statistical comparison by two-sided Mann–Whitney U test with BCa bootstrap confidence intervals. Spearman rank correlations assessed potential confounders (pLDDT, disorder fraction, radius of gyration). AlphaFold confidence sensitivity analysis stratified proteins into high-confidence (pLDDT ≥ 70, $n = 11$) and low-confidence ($n = 11$) subsets.

GWAS enrichment analysis

AIS risk genes from Gorman *et al.*[?] and associated meta-analyses. Each gene product COI computed as above. Fisher's exact test for enrichment of high-COI proteins (top quartile, COI > 3.1) among risk loci versus random genome expectation.

Statistical methods

All analyses were performed in Python 3.11 (scipy 1.11, scikit-learn 1.3, numpy 1.24, statsmodels 0.14). All tests are two-sided unless otherwise noted. Multiple comparison corrections use the Bonferroni method where applicable. Effect sizes follow Cohen's conventions. Bootstrap resampling used $n = 10{,}000$ iterations. Random seed: 42 for all stochastic procedures. Power analysis: $n = 35$ AIS patients needed for 80% power ($\alpha = 0.05$) to detect $r \approx 0.46$. Full statistical results are provided in Supplementary Table S1.

Data availability

Simulation code (PyElastica), AlphaFold analysis scripts, cross-species scaling data, and protein COI datasets for all 47 proteins are available in the `spinalmodes` public repository (`https://github.com/spinalmodes`).

Code availability

All custom analysis code is available at `https://github.com/spinalmodes`.

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Author contributions

S.K.S. conceived the framework, performed the analyses, wrote the manuscript, and prepared all figures.

Competing interests

The author declares no competing interests.

Additional information

Supplementary Information is available for this paper.

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Extended Data Table 1 | Conformational Ordering Index values for AIS-implicated proteins.

Functional Tier	Protein	UniProt	Residues	COI	Disorder	Role
Mechanosensors	Vimentin (VIM)†	P08670	466	7.47	0.12	IF strain gauge
	Integrin-β1†	P05556	798	4.83	0.15	ECM–cytoskeleton coupling
	Piezo2†	Q9H5I5	2,822	3.91	0.22	Proprioceptive ion channel
	Lamin A/C†	P02545	664	3.67	0.31	Nuclear mechanosensor
	YAP1†	P46937	504	2.91	0.45	Mechanotransduction effector
	Talin-1†	Q9Y490	2,541	3.44	0.18	Focal adhesion force sensor
	Filamin A†	P21333	2,647	4.12	0.14	Actin cross-linker
Structural	Collagen I (α1)	P02452	1,464	5.89	0.08	Load-bearing matrix
	Collagen II (α1)	P02458	1,487	5.71	0.09	Disc matrix
	Fibrillin-1	P35555	2,871	4.23	0.11	Elastic fibre scaffold
	Aggrecan	P16112	2,415	2.88	0.38	Disc proteoglycan
	Elastin	P15502	786	1.34	0.71	Elastic fibre protein
	Fibronectin†	P02751	2,386	3.56	0.19	ECM adhesion
Metabolic Regulators	PPARGC1A (PGC-1α)	Q9UBK2	798	2.19	0.62	Mitochondrial biogenesis
	SIRT1	Q96EB6	747	1.92	0.41	NAD^+ energy sensor
	AMPK-α1	Q13131	559	1.78	0.33	Energy homeostasis kinase
	AMPK-β1	Q9Y478	270	1.45	0.48	AMPK regulatory subunit
	ARNTL (Bmal1)	O00327	626	3.31	0.37	Circadian clock core
	GHR	P10912	638	1.67	0.52	Growth hormone receptor

Twenty-five representative proteins from the full 47-protein cohort are shown (complete dataset available at `https://github.com/spinalmodes`). Protein structures retrieved from AlphaFold Protein Structure Database v4. COI $= (\lambda_{\max} - \lambda_{\min})/\bar{\lambda} \times \ln(N)$ computed from C_α gyration tensor eigenvalues. Disorder fraction: proportion of residues with pLDDT < 70. Proteins ordered by COI within each functional tier. GWAS† indicates AIS genome-wide association study risk locus. Demand-side mean COI $= 3.74 \pm 0.81$ ($n = 13$); supply-side mean COI $= 1.74 \pm 0.43$ ($n = 12$); gap $p < 0.001$, Mann–Whitney U.

Extended Data Table 2 | Cross-species bio-gravitational scaling analysis.

Species	Mass (kg)	Spine L (m)	EI ($\mathrm{N\,m^2}$)	Bg	Regime
Mouse	0.025	0.04	3.8×10^{-5}	0.110	Passive stable
Rat	0.30	0.10	4.2×10^{-3}	0.047	Passive stable
Rabbit	2.5	0.20	0.12	0.025	Transitional
Cat	4.0	0.30	0.28	0.021	Transitional
Dog	20	0.45	2.1	0.012	Active required
Human (child)	30	0.35	1.8	0.015	Allometric Trap
Human (adult)	70	0.50	5.5	0.0076	Allometric Trap
Horse	500	0.80	85	0.011	Active required
Elephant	5,000	1.20	450	0.0051	Active required

Bio-gravitational number $Bg = EI/(MgL^2)$ computed from published biomechanics data. Log-log regression slope $= -0.282$ ($R^2 = 0.744$, $p = 0.0013$); BCa 95% CI: $[-0.405, -0.170]$; permutation $p = 0.0014$. Metabolic scaling prediction (-0.25) falls within CI ($t(8) = 0.55$, $p = 0.60$). Bayes factor ≈ 290 versus null model. Human children show $Bg_{\mathrm{child}}/Bg_{\mathrm{adult}} = 1.99$, confirming maximal vulnerability during the adolescent growth window.

Supplementary Note 1 | The Geodesic Deviation Analogy

The following is a formal analogy, not a literal application of general relativity, included to establish precise physical correspondence.

In general relativity, a test mass in free fall follows a spacetime geodesic[?] . A standing organism deviates from this geodesic with proper acceleration a^{μ} proportional to the metabolic power expended by muscle contraction. The biological stress-energy tensor may be written:

$$T_{\mu\nu}^{\mathrm{bio}} = \rho_{\mathrm{ATP}}\, u_{\mu} u_{\nu} + p_{\mathrm{mito}}\, h_{\mu\nu}$$

where ρ_{ATP} is the metabolic energy density, p_{mito} is the "metabolic pressure" (mitochondrial power density), and $h_{\mu\nu} = g_{\mu\nu} + u_{\mu} u_{\nu}$ is the spatial projection tensor. When metabolic buckling occurs — $T_{\mu\nu}^{\mathrm{bio}}$ falls below the threshold — the spine adopts the configuration of minimum proper acceleration: the helical scoliotic curve. In this precise sense, AIS represents a **biological geodesic relaxation**: the organism trades geometric perfection for thermodynamic survival.